

SWIFT TOWERS AS OPEN RESEARCH LAB: THE EXPERIENCE OF A NETWORK OF NESTING SITES IN NW ITALY

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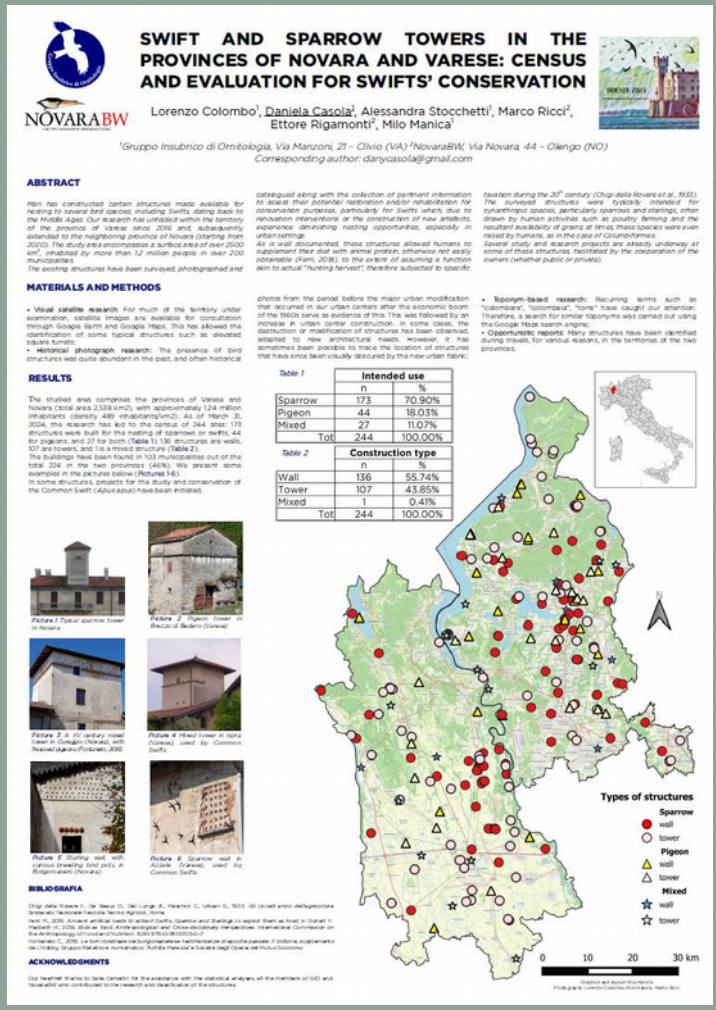
31.05.2024

Swift towers in NW Italy



We are studying since 2018 some structures used by Common swifts *Apus apus*

We mapped them in NW Italy (some of them are mapped in a poster!)



SWIFT AND SPARROW TOWERS IN THE PROVINCES OF NOVARA AND VARESE: CENSUS AND EVALUATION FOR SWIFTS' CONSERVATION

NOVARABW

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ABSTRACT

MATERIALS AND METHODS

RESULTS

Table 1

	Intended use	
	n	%
Sparrow	173	70.90%
Pigeon	44	18.03%
Mixed	27	11.07%
Tot	244	100.00%

Table 2

	Construction type	
	n	%
Wall	136	55.74%
Tower	107	43.85%
Mixed	1	0.41%
Tot	244	100.00%

Types of structures
Sparrow
wall
tower
Pigeon
wall
tower
Mixed
wall
tower

0 10 20 30 km

BIBLIOGRAFIA

ACKNOWLEDGMENTS

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Studied sites



Three structures:
- Jerago con Orago (VA): 105 cells
- Azzate (VA): 200 cells
- Cuggiono (MI): 299 cells

One public building
Two private buildings







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Materials and methods



Monitor weekly during the breeding season (three colonies)

Cork coaster as nesting attractants (two colonies)

5 IoT tracking devices on breeding individuals (one colony)



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Materials and methods



Scheda rilievo

Data				Meteo	Rilevatori	ID uscita
gg	mm	aa	ora			
29	04	24	9:36	nuvolosità: 60% precipitazioni: 0 vento: 0	Claudia Iadanza	02/2024

A-E SOPRA STRADA SX

A1	A2	A3	A4	A5	A6	A7	A8	A9
		U: 3 / A: 1						

B1	B2	B3	B4	B5	B6	B7	B8	B9

C1	C2	C3	C4	C5	C6	C7	C8	C9
U: 3								

D1	D2	D3	D4	D5	D6	D7	D8	D9
							U: 3	

E1	E2	E3	E4	E5	E6	E7	E8	E9

A-E SOPRA STRADA DX

A10	A11	A12	A13	A14	A15	A16	A17
U: 4			U: 3				

B10	B11	B12	B13	B14	B15	B16	B17
U: 3					U: 3	U: 3	

C10	C11	C12	C13	C14	C15	C16	C17
			U: 2 fuori dal nido				

D10	D11	D12	D13	D14	D15	D16	D17

E10	E11	E12	E13	E14	E15	E16	E17
					A: 1 morto, no anello		



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Materials and methods



We test the effect on reproductive phenology and population parameters of meteorological data in General Linear Models (Gaussian)

Attractant and (re-)colonization: Generalized Linear Models with binomial error distribution (dependent variable: 0 non-used; 1 used)

Tracking system is based on SigFox IoT network. Daily travel distances and movement metrics calculated in R

Results 1/3 – phenology

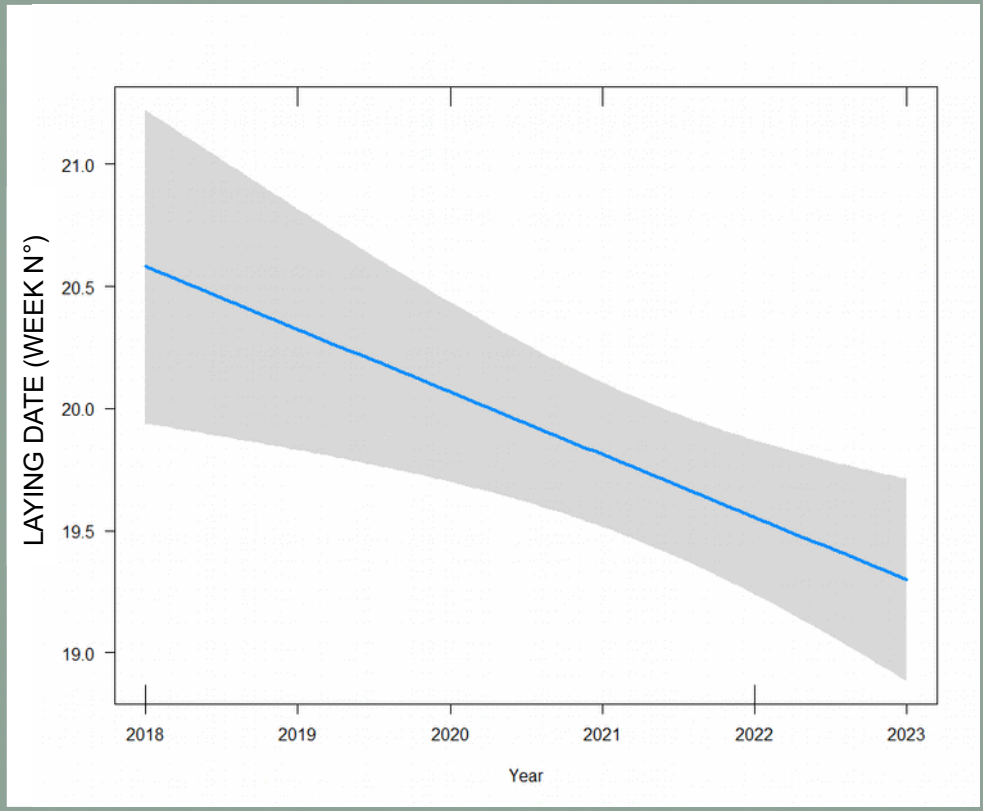


EARLIER DEPOSITION

Impressive trend of earlier deposition dates (1 week in 6 years)!

This is likely to be a direct effect of spring warming, in line with several publications

NOTE: the frequency of data collection (weekly) do not allow to observe finer advancement in laying date (i.e. below 1 week). It is noteworthy that we found in any case such a strong effect!



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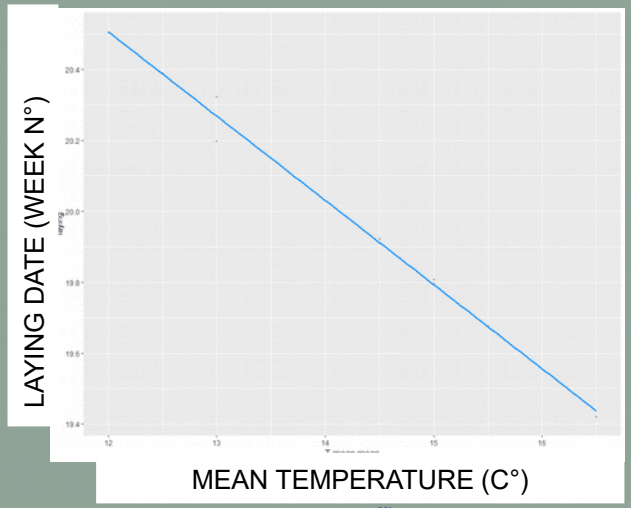
Results 1/3 – phenology

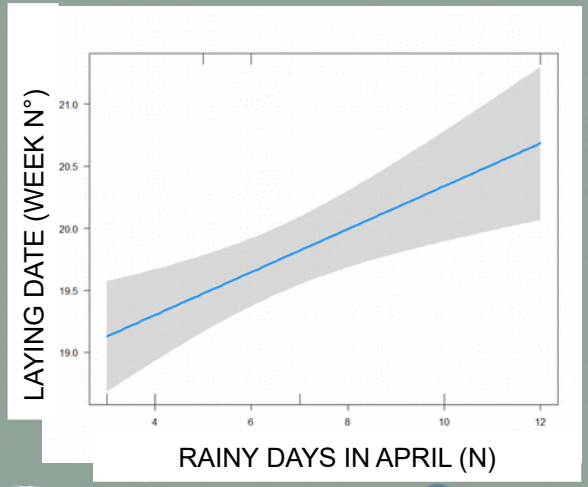


TEMPERATURE AND RAIN EFFECTS

- The medium temperature of April and May influenced eggs laying: the warmer the years, the earlier swifts lay their eggs. It is not a spurious result because the medium temperature throughout the years does not increase.
- The more often it rains, the later swifts lay their eggs.

However, accordingly to our models, temperature is the factor that most influences egg deposition.





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Results 2/3 – cork coaster



OLD NESTS vs CORK COASTER and EMPTY CELLS

Cavities with old natural nest are preferred over empty ones, but also over those with unused coasters.

Factor	Estimate	SE	z	p
Intercept	2.53	0.74	3.43	<0.001
Empty cavities VS old nests	-6.18	1.02	-6.03	<0.001
Coark coasters VS old nests	-5.00	0.85	-5.89	<0.001
Cuggiono VS Azzate	-1.41	0.8	-1.75	0.081
Cork coaster * Locality	2.70	1.14	2.38	0.017
Old nest * Locality	2.03	0.97	2.10	0.036

Results 2/3 – cork coaster



CORK COASTER vs EMPTY CELLS

Since the two colonies have a different histories (in Cuggiono the colony was almost extinct due to bad maintenance), we verified only in this colony whether the positioning of cork coasters could accelerate recolonization.

Factor	Estimate	SE	z	p
Intercept	-0.56	0.17	-3.27	0.002
Coark coasters VS Empty cavities	0.63	0.24	2.65	0.008

We found that in this colony, cells with cork coasters had a significantly higher probability of being occupied than empty cells.

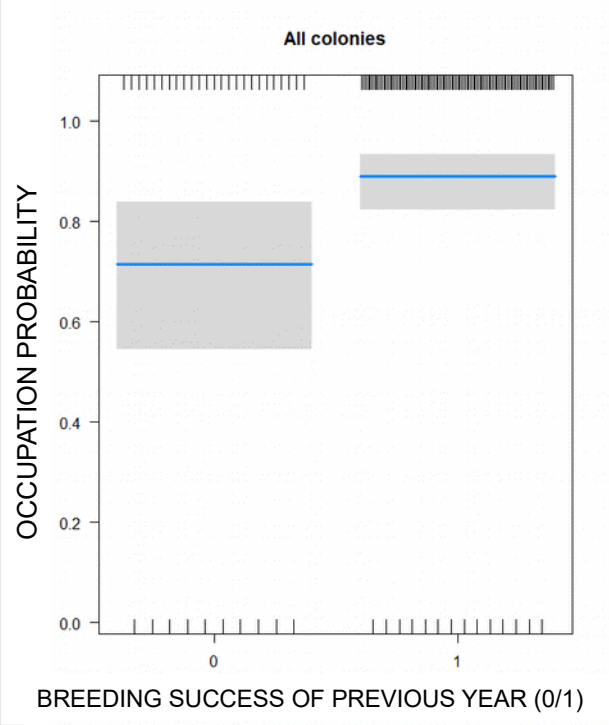
Therefore, in 'swift walls/towers' with many empty cells, the use of cork coasters is an effective accelerator of colonization.

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Results 2/3 – cork coaster



PROBABILITY OF CELL OCCUPANCY



The cells where successful reproduction occurred (in the previous year) are significantly more occupied than those where reproduction was unsuccessful

(considering only the cells where nesting occurred in the previous year)

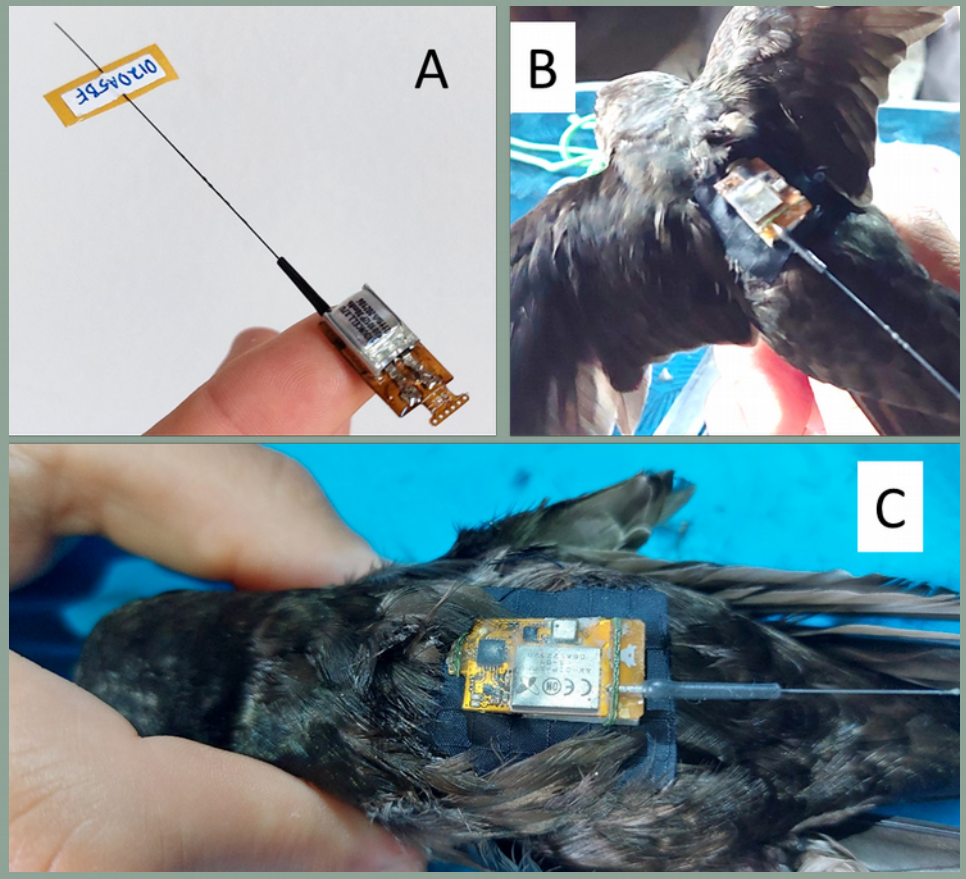
Strong effect $p<0.001$ (binomial GLM)

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Results 3/3 – Tracking IoT



DEVICES AND MIGRATION



- Quality of the data is acceptable
- Inaccuracy in the locations and their frequency (12h) still prevent the possibility of using these for a detailed study of the foraging ecology
- Devices were deployed without a harness and were well tolerated by the birds, suggesting this may be a common way to deploy swifts in the future
- **Birds were back in 2024 safe and without the devices, so they successfully lost them as predicted**

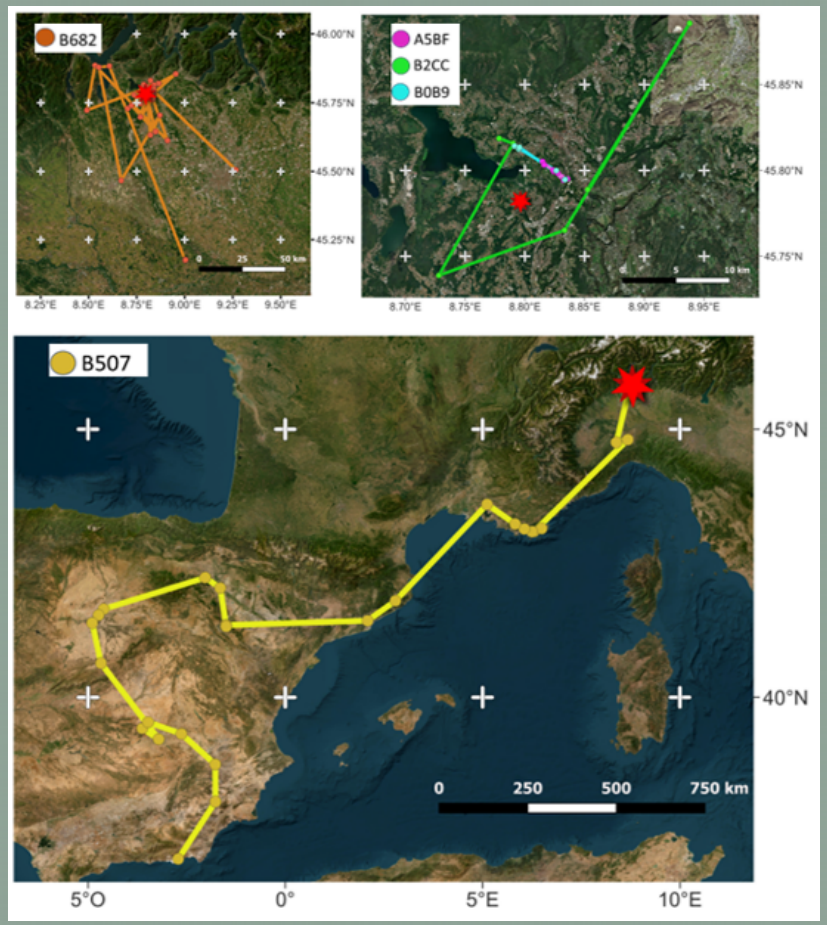
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Results 3/3 – Tracking IoT



DEVICES AND MIGRATION

- All of the five devices successfully transmitted data (92 valid locations: 62 were accompanied by the estimation of the location error)
- On average, the devices collected 19 locations each (min 6, max 45), with an overall average error of 7.44 km (max: 15.6 km; min: 3.4 km; sd: 3.55 km)
- 4 devices collected information on movements likely related to foraging, close to the colony
- 1 individual left the colony and started post-nuptial migration (tracked July 1 - July 16)



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Conclusion



- Swifts' breeding phenology is quickly responding to global change, most likely due to spring warming
- Cork coasters may be an effective tool to accelerate re-colonization of (near to) deserted swift walls/towers
- In old, well established colonies, the effect of cork coasters in favouring occupation is positive but negligible
- Remote tracking of swifts is now reliable, with further technical improvement of IoT devices may soon will gather data of higher quality
- Swift walls/towers are a valid open laboratory for training and involving new volunteers

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Thanks for your attention









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